NASA SPACE APPS 2026 // EARTH SYSTEM TREND DETECTIVE

The Milan Heat Case

Mapping Urban Climate Inequality Across 6,079 Census Tracts (2000–2023)

3-Min Video: youtu.be/7MqjsmNJB54 milan-heat-explorer.streamlit.app github.com/marybilik/milan-heat-explorer

01 // THE URBAN HEAT PARADOX

Extreme Micro-Scale Thermal Inequality

6.87°C

Surface temperature gap within 300 meters

Historic concrete core vs. Parco Sempione in Summer 2023 Landsat 8/9 telemetry (peaking at 54°C LST).

7.6x Extreme Heat Exposure

19.2% of vulnerable tracts exceed 50°C vs. 2.5% affluent.

Accelerating Decadal Gap

Q4 warms +0.858°C/dec vs. +0.825°C/dec in Q1 ($p < 0.0001$).

571,146 Residents at Risk

40.8% of Milan lives in heat or pollution danger zones.

02 // ORBITAL & GROUND DATA FUSION

Multi-Sensor NASA & Partner Architecture

Source / Agency	Dataset / Catalog ID	Scale	Analytical Role
NASA / USGS	Landsat 8 & 9 C2 L2 (LST)	30 m (2014–2023)	Street-level thermal gaps & decadal slopes
NASA LP DAAC	MODIS MOD11A2 & MOD13A2	1 km (2000–2023)	24-year LST & NDVI climate baseline
ESA Copernicus	Sentinel-5P Tropospheric NO_2	5.5×3.5 km (2023)	Air pollution & Heat Double Burden
EC JRC & Zenodo	GHSL Pop 2020 & Milan SVI	100 m / 6,079 Tracts	Population-weighted risk & 8 census metrics
OSM & Google Earth	OSM Vectors & 3D Basemaps	Sub-meter 3D	Industrial & rail residual verification

03 // EMPIRICAL INVESTIGATION FINDINGS

Four Hidden Drivers of Urban Heat Risk

The Physics Illusion

Concrete cover explains 57.5% of LST variance.

The Loneliness Trap

One-person households add +0.083°C per SD.

The Industrial Ovens

59.2% of extreme heat anomalies hit logistics hubs.

The Invisible Majority

49.3% of vulnerable residents live in orange zones.

04 // ECONOMETRIC & SPATIAL ML RIGOR

Verified Against Spatial Bias & Tail Risk

2.07%

5×5 Spatial Block CV MAPE (300-Tree Random Forest)

10×

SVI tail-risk escalation at Q95 quantile regression

4.3×

Industrial & rail enrichment in +2°C residual tracts

---

1. Benjamini-Hochberg FDR

Eliminates false demographic correlations.

2. Spatial Lag Control

Corrects Moran's I = 0.74 autocorrelation.

3. GWR & Bivariate LISA

Maps 589 Red & 687 Orange target tracts.

05 // INTERACTIVE TOOL & POLICY IMPACT

From 5-Phase Detective Game to Civic Action

5-Phase Streamlit Investigation

Phase 1 — Crime Scene: 300 m Duomo vs. Park gap.

Phase 2 — Time Bomb: +2.35°C summer amplification.

Phase 3 — Invisible Enemy: OSM industrial anomaly layer.

Phase 4 — Hotspots: Local GWR coefficient explorer.

Phase 5 — Invisible Majority: Orange-zone LISA discovery.

3 Evidence-Based Interventions

Cool Roofs (-0.36°C): Outperform trees (+0.04°C) in saturated Q4 tracts.

Industrial Buffers: High-albedo roofs across 253 logistics hotspots.

Neighbor Registry: Proactive outreach in 687 orange isolation tracts.

OPEN SCIENCE // READY FOR CIVIC DEPLOYMENT

Every Degree Has a Geography. Now Milan Can Map It.

Explore the live spatial simulator, inspect the 3-minute walkthrough, or reproduce the full econometric pipeline on GitHub.

3-MINUTE VIDEO DEMO
youtu.be/7MqjsmNJB54

LIVE STREAMLIT TOOL
milan-heat-explorer.streamlit.app

OPEN SOURCE REPO
github.com/marybilik/milan-heat-explorer